

in the heat of conflict was a foot soldier's essential skill. Before he could fire, he had to take out a paper cartridge containing the minié ball and powder charge; rip open the cartridge with his teeth; ram powder and ball down the barrel with his ramrod; and take a percussion cap from its box and place it in the firing mechanism. Yet this apparently clumsy, muzzle-loaded weapon marked a vital evolution in warfare. An experienced man might fire his Springfield three times a minute, its rifled, or grooved, barrel offering improved accuracy and an effective range of 300–400yd (274–366m)—three times that of a smoothbore flintlock. Under pressure, men would make mistakes, leaving the ramrod in the barrel or even forgetting to load the powder and ball. But a rifle musket, correctly used, was a reliable arm. In defense, its range and rate of fire presented a serious problem for an attacking force that had to cover open ground in a frontal assault.

5 The factor by which firepower was improved when flintlock muskets were replaced by rifle muskets, increasing both the effective firing range of Civil War infantry and the rate of fire.